

--	--	--	--	--	--	--	--

B.Tech. Degree VI Semester Special Supplementary Examination July 2019

CS 15 - 1603 CRYPTOGRAPHY AND NETWORK SECURITY (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Differentiate between cryptography and cryptanalysis.
- (b) What are the different types of viruses? Discuss its characteristics.
- (c) What is meant by message digest? Give an example.
- (d) Describe the Miller-Rabin algorithm and test the primality of 23.
- (e) Distinguish conventional encryption with public key encryption?
- (f) What are different modes of operation in block cipher?
- (g) Discuss different protocols used in SSL.
- (h) Discuss the Feistel Cipher with three rounds.

Encrypting the message "Hide the gold in the tree stump" Using Affine Cipher ($a=5, b=8$).

Describe the different types of security attacks in networks.

PART B

(4 × 10 = 40)

- II. (a) With a neat sketch discuss the functioning of network security model. (5)
- (b) Encrypt the following using play fair cipher using the keyword MONARCHY. (5)
"SWARAJ IS MY BIRTH RIGHT". Use X for blank spaces.

OR

- III. Explain any three substitution techniques and two transposition techniques with suitable example? (10)

- IV. Discuss various steps in the DES Encryption algorithm with neat diagrams. (10)

OR

- V. Explain AES Encryption algorithm in detail with necessary diagrams. (10)

- VI. (a) Briefly explain Diffie- Hellman Key exchange with suitable example. (5)
- (b) Distinguish between SHA1 and MD5 hash functions. (5)

OR

- VII. Write notes on the following: (10)
 - (i) RSA Cryptosystem
 - (ii) ElGamal Scheme

- VIII. Discuss how firewalls help in the establishing a security framework for an organization. (10)

OR

- IX. Write short notes on the following: (10)
 - (i) Kerberos
 - (ii) IP security
 - (iii) Digital Signature Standards
